جامعة الملك سعود
King Saud University

Nakhlah

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Introduction

A. Project Idea

Nakhlah is a mobile application designed to help anyone interested in dates whether tourists, consumers, farmers, or enthusiasts to instantly identify date varieties through AI-powered image recognition.

By simply taking a photo of a date, the app provides:

- Date Type Identification (e.g., Medjool, Deglet Noor, Ajwa, Sukkari)
- Origin Information (geographical source and cultivation region)
- Nutritional Information (calories, sugar content, fiber, vitamins, minerals)

The app serves as an educational tool and practical guide for date lovers worldwide, promoting awareness of date varieties and their health benefits. It supports both English and Arabic languages to serve local and international users.

B. Stakeholders

- Tourists and Visitors: People exploring Saudi Arabia and Middle Eastern culture who want to learn about dates
- Date Consumers: General public interested in nutrition and health benefits
- Date Enthusiasts: People passionate about dates and their varieties
- Farmers and Retailers: Those in the date industry seeking variety information
- Health-Conscious Users: Individuals tracking nutritional intake

C. Glossary

Term	Definition
User	Any registered individual who uses the app to scan and identify dates
Date Type	The variety or cultivar of the date (e.g., Medjool, Ajwa, Deglet Noor, Sukkari)
Origin	The geographical region or country where the date variety is primarily grown
Nutrition Profile	Detailed nutritional information including calories, sugar, fiber, vitamins, and minerals
Scan History	Record of all dates previously scanned by a user
AI Model	Custom-trained machine learning model that identifies date varieties from images
Date Database	Backend collection of date varieties with their characteristics, origins, and nutritional data
Profile	User account containing personal preferences and scanning history

Product Backlog

A. System Features

Actor	ID	System Features
USER	1	User can sign up using name, email, and password.
	2	User can sign in using email and password.
	3	User can reset password via email.
	4	User can manage and update their profile.
	5	User can log out from their account.
	6	User can capture or upload a photo of a date.
	7	User can view identification result (date type, confidence score).
	8	User can view origin information of identified date.
	9	User can view nutritional information (calories, sugar, fiber, vitamins).
	10	User can switch between English and Arabic languages.
	11	User can view scan history of previously identified dates.
	12	User can save favorite date types to their profile.

USER	13	User can browse all date varieties in the database.
	14	User can search for specific date types.
	15	User can delete items from scan history.
	16	User can browse the Nakhlah Marketplace of trusted sellers.
	17	User can view seller profile, products, and contact info.
	18	User can filter marketplace by date variety, region, or seller rating.
	19	User can share scan results with friends via social media / messaging.
	20	User can compare nutritional values of two or more date varieties .
SYSTEM	21	System identifies date type using AI model with confidence score.
	22	System retrieves and displays origin information from database.
	23	System retrieves and displays nutritional data from database.
	24	System stores scan results and user history in Firestore.
	25	System works offline with pre-loaded AI model.
	26	System displays verified seller badges on marketplace listings.

B. Technical Features

ID	Feature	Priority
1	Install Visual Studio Code	MUST
2	Installation of Git	
3	Create GitHub repository	
4	Configure Flutter environment	
5	Install Firebase	
6	Create Figma workspace	
7	Set up Python + PyTorch environment	
8	Configure Android emulator	

C. Knowledge Acquisition

ID	Feature	Priority
1	Learn Flutter development	MUST
2	Learn Firebase (Auth, Firestore, Storage)	
3	Learn Git commands and version control	
4	Learn UI/UX design in Figma	
5	Learn PyTorch for AI model training	
6	Learn image classification and CNNs	
7	Learn dataset collection and labeling	

B. User Stories

ID	Title	User Story	Priority	Points	Category
1	Sign Up	As a new user, I want to sign up using my name, email, and password, so I can create an account and access the app.	MUST	3	Auth
2	Sign In	As a user, I want to sign in with my email and password so I can access my account and scan history.	MUST	2	Auth
3	Reset Password	As a user, I want to reset my password via email so I can regain access if I forget it.	MUST	2	Auth
4	Manage Profile	As a user, I want to manage and update my profile (name, email, preferences) so my information stays accurate.	MUST	3	Profile
5	Log Out	As a user, I want to log out so I can keep my account secure.	MUST	1	Auth
6	Capture Photo	As a user, I want to capture or upload a photo of a date so that the app can identify it.	MUST	5	Core
7	Language Toggle	As a user, I want to switch between English and Arabic so I can use the app in my preferred language.	MUST	3	UX
8	AI Identify Date	As a user, I want the AI to identify the date type from my photo so I can learn what variety it is.	MUST	8	AI
9	Confidence Score	As a user, I want to see how confident the AI is in its identification so I know how reliable the result is.	SHOULD	2	AI
10	View Origin	As a user, I want to see the origin of the identified date so I know where it's typically grown.	MUST	3	Info
11	View Nutrition	As a user, I want to see nutritional information (calories, sugar, fiber, vitamins) so I can make informed dietary choices.	MUST	3	Info
12	Scan History	As a user, I want to view my previous scans so I can review dates I've identified before.	SHOULD	3	History
13	Delete History	As a user, I want to delete items from my scan history so I can remove unwanted records.	COULD	2	History
14	Browse Database	As a user, I want to browse all date types in the database so I can explore different varieties without scanning.	SHOULD	3	Discovery
15	Search Database	As a user, I want to search for specific date varieties by name so I can quickly find information.	SHOULD	2	Discovery

16	Save Favorites	As a user, I want to save favorite date types so I can quickly access information about dates I like.	SHOULD	2	Profile
17	Compare Dates	As a user, I want to compare nutritional values between different dates so I can make better choices.	COULD	5	Info
18	Share Results	As a user, I want to share my scan results so I can share interesting date varieties with friends.	COULD	3	Social
19	Browse Marketplace	As a user, I want to browse a marketplace of trusted date sellers so I can discover and buy quality dates.	MUST	8	Marketplace
20	View Seller Profile	As a user, I want to view a seller's profile, products, and contact info so I can make an informed purchase decision.	MUST	5	Marketplace
21	Filter Marketplace	As a user, I want to filter sellers by date variety, region, or rating so I can find what I'm looking for quickly.	SHOULD	5	Marketplace
22	Seller Badge	As a user, I want to see a verified badge on trusted sellers so I know they are approved by Nakhlah.	SHOULD	2	Marketplace
23	Similar Dates	As a user, I want the system to suggest similar date varieties after scanning so I can discover new types I might like.	COULD	5	AI
24	Offline Mode	As a user, I want basic identification to work offline so I can use the app without an internet connection.	WOULD	8	Core
25	Date Ripeness	As a user, I want the AI to indicate the ripeness stage of a date (Khalasl, Rutab, Tamr) so I know when it's best to eat.	COULD	8	AI
26	Nutritional Goal Track	As a user, I want to set a daily sugar or calorie goal and track my date consumption so I can manage my diet.	WOULD	8	Health

Non-Functional Properties

ID	Property	Why It's Critical
1	Performance	AI inference must complete within 3 seconds. App should load within 2 seconds. Users expect instant results and smooth scanning experience.
2	Accuracy	AI model should achieve minimum 85% accuracy on date identification. Confidence scores must be calibrated and reliable for user trust.
3	Security	User passwords must be hashed and encrypted. Authentication tokens secured. Secure API communication (HTTPS). User data privacy protected per Firebase security rules.
4	Usability	Intuitive UI requiring no tutorial. Clear visual feedback for all actions (camera, scanning, results). Accessible design (font sizes, contrast). Supports both LTR (English) and RTL (Arabic) layouts.
5	Localization	Full bilingual support (English/Arabic). Proper RTL layout for Arabic. Culturally appropriate content. Date formats and numbers localized.
6	Reliability	Image upload and AI inference must work consistently. Scan history data must be persisted reliably. No data loss during network issues (offline queue).

Infrastructure Readiness

ID	Type	Name	Status	Description
1	Software	Flutter	Available	SDK for building the mobile application interface.
2	Software	Dart	Available	Programming language used with Flutter.
3	Software	Android Emulator	Available	Used to test and run the mobile application.
4	Software	Python	Available	Detailed nutritional information including calories, sugar, fiber, vitamins, and minerals
5	Software	PyTorch	Required	Framework for building and training AI models.
6	Software	NumPy	Required	Library for numerical computations.
7	Software	Pandas	Required	Library for data processing and analysis.
8	Software	Laptop / PC	Available	Device for development, testing, and documentation.
9	Network	Wi-Fi Connection	Available	Internet access for development and cloud services.

Environment Specification

ID	Type	Name	Status	Description
1	Software	VS Code	Available	Code editor for developing frontend and backend components.
2	Software	Trello	Available	Task and project management tool for collaboration and tracking
3	Software	Figma	Required	UI/UX design and prototyping tool.
4	Software	Discord	Available	Team communication and meetings.
5	Software	WhatsApp	Available	Quick communication and updates.
6	Software	Git	Required	Version control system for tracking code changes.
7	Software	GitHub	Available	Remote repository hosting and collaboration platform.
8	Software	Google Colab	Required	Cloud environment for AI model training and experiments.

Team Readiness

#	Name	Skill	Status
1	Mansour Abahussain	AI Specialist, Fullstack	To Be Trained
2	Mohammad Alfarraj	UI/UX, Fullstack	To Be Trained
3	Abderrahmane Benfatah	AI Specialist, Fullstack	To Be Trained
4	Ahmed Albabtain	UI/UX, Fullstack	To Be Trained
5	Omar Ramadhan	Testing, UI/UX, Fullstack	To Be Trained

Training

Name	Skill	Related To	Person to be Trained	Source	Duration
Flutter	Programming	Mobile App Development	Mohammad , Omar and Ahmed	YouTube/ Udemy	One Week
Figma	Design	UI/UX	Mohammad , Omar and Ahmed	YouTube	Three Days
Trello	Management	Project Management	All members	YouTube	One Hour
FireBase	Programming	Backend , Database	Abderrahmane and Mansour	YouTube	One Week
Python	Programming	Pytorch, Colab, Numpy Pandas	Abderrahmane and Mansour	YouTube	One Week

System Architecture Design

We adopted a Multi-Tier Architecture to separate the system into independent layers. Each layer has a specific responsibility, which improves organization, maintainability, scalability, and system clarity.

Layer 1 — Presentation Layer (Mobile Application)

Purpose :

Provides the user interface for interacting with the system.

Technologies :

Flutter with Firebase Authentication and Firebase Storage.

Functions / Responsibilities :

Handles user login and registration, captures images, uploads images to cloud storage, sends analysis requests to the backend API, and displays prediction results and user history.

Layer 2 — Application Layer (API Backend)

Purpose :

Acts as the intermediary between the mobile application and the AI model.

Technologies :

FastAPI framework using Python.

Functions / Responsibilities :

Validates user authentication, receives image analysis requests, forwards them to the AI Inference Layer, stores prediction results in the database, and returns structured responses to the mobile application.

Layer 3 — AI Inference Layer

Purpose :
Executes the trained machine learning model and generates predictions.
Technologies :
PyTorch deployed inside a Docker container on a GPU-enabled server.
Functions / Responsibilities :
Loads the trained model, preprocesses input images, performs inference, postprocesses prediction outputs, and returns structured JSON results to the Application Layer.

Layer 4 — Data Layer

Purpose :
Manages persistent storage of system data and user information.
Technologies :
Firebase Storage and Firestore database.
Functions / Responsibilities :
Stores uploaded images, saves prediction results, maintains user scan history, and records timestamps and related metadata.

Layer 1 — Presentation Layer (Mobile Application)
↓
Layer 2 — Application Layer (API Backend)
↓
Layer 3 — AI Inference Layer
↓
Layer 4 — Data Layer

Project Management Plan

1. Project Overview

Nakhlah is a Flutter-based mobile app that uses AI to identify date varieties from images and provide nutritional information and calorie counts. The project integrates mobile development, Firebase backend, and a machine learning classification model.

2. Development Methodology

The team will follow Agile Scrum:

- 2-week sprints
- Sprint planning, review, and retrospective
- GitHub for task tracking and version control

3. Major Phases

1. **Planning & Design**

- Define requirements
- Design UI
- Prepare dataset

2. **AI Development**

- Train classification model
- Test accuracy (target ≥ 85%)

3. **App Development**

- Build Flutter UI
- Integrate Firebase
- Implement camera & upload

4. **Integration & Testing**

- Connect AI model to app
- Perform functional testing
- Fix bugs and optimize performance

Sprint Plan & Release Roadmap

Sprint 1 — Foundation (Weeks 1–2) | 20 Points

ID	Title	Acceptance Criteria	Priority	Points
1	**Sign Up**	Email format validated; duplicate email blocked; password min 8 chars	MUST	3
2	**Sign In**	Valid credentials grant access; invalid shows error	MUST	2
3	**Reset Password**	Reset email sent within 60 s; link expires after 1 h	MUST	3
4	**Manage Profile**	User can update name/email; changes persisted in Firestore	MUST	3
5	**Log Out**	Session terminated; user redirected to login screen	MUST	1
6	**Capture Photo**	Camera/gallery picker works on Android; preview shown before analysis	MUST	5
7	**Language Toggle**	Full RTL layout for Arabic; all strings translated	MUST	3

Sprint 2 — AI Core & Information (Weeks 3–4) | 26 Points

ID	Title	Acceptance Criteria	Priority	Points
8	**AI Identify Date**	Model returns result in ≤3 s; accuracy ≥85% on test set	MUST	8
9	**Confidence Score**	Percentage shown with color indicator (green/yellow/red)	SHOULD	2
10	**View Origin**	Region, country, and cultivation notes displayed after scan	MUST	3
11	**View Nutrition**	Calories, sugar, fiber, vitamins shown per 100 g serving	MUST	3
12	**Scan History**	Last 50 scans listed with date, thumbnail, identified type	SHOULD	3
13	**Delete History**	Swipe-to-delete or select-and-delete; confirmation dialog shown	COULD	2
14	**Browse Database**	All varieties listed alphabetically; photos included	SHOULD	3

Sprint 3 — Marketplace & Engagement (Weeks 5–6) | 30 Points

ID	Title	Acceptance Criteria	Priority	Points
15	**Search Database**	Real-time filter by name in English and Arabic	SHOULD	2
16	**Save Favorites**	Heart icon on detail screen; persisted to user profile	SHOULD	2
17	**Compare Dates**	Side-by-side nutritional table for up to 3 varieties	COULD	5
18	**Share Results**	Generates shareable card image; supports WhatsApp, Instagram	COULD	3
19	**Browse Marketplace**	Grid of trusted seller cards with logo, name, rating, specialty	MUST	8
20	**View Seller Profile**	Full seller page: bio, products, region, contact/WhatsApp link	MUST	5
21	**Filter Marketplace**	Chips for date variety, region, and star rating; real-time filter	SHOULD	5
22	**Seller Badge**	Verified green checkmark on all approved seller cards	SHOULD	2

Sprint 4 — Advanced Features & Polish (Weeks 7–8) | 29 Points

ID	Title	Acceptance Criteria	Priority	Points
23	**Similar Dates**	3 related varieties suggested based on scan result	COULD	5
24	**Offline Mode**	AI inference runs locally; cached DB data accessible without internet	WOULD	8
25	**Date Ripeness**	Ripeness stage (Khalal/Rutab/Tamr) displayed with scan result	COULD	8
26	**Nutritional Goal Track**	Daily sugar/calorie goal set; intake tracked per scan session	WOULD	8

4. Roles & Responsibilities

- AI Lead – Model training and testing
- Flutter Developer(s) – Frontend implementation
- Backend Developer – Firebase & database
- Project Manager Sprint planning, tracking progress, risk management, documentation

5. Risk Management

Risk	Risk Meaning	Severity	Response Strategy
Low AI Accuracy	The model fails to correctly classify date varieties due to limited or poor-quality dataset.	High	Improve dataset quality, use data augmentation, apply transfer learning, and retrain model.
Integration Issues	Difficulty connecting the ML model with the Flutter app or Firebase backend.	Medium	Perform early integration testing and modular development.
Time Constraints	Sprint delays or unfinished features before final deadline.	High	Strict sprint planning, prioritize core features, and monitor progress weekly.
Dataset Limitations	Insufficient images of certain date varieties affecting model performance.	Medium	Collect diverse images from multiple sources and balance dataset.
Team Availability	Team members unavailable due to exams or other coursework.	Medium	Distribute tasks clearly and maintain shared documentation.
Development Environment Setup	Difficulty installing and configuring Flutter, Android SDK, emulators, or Firebase tools on different machines.	Medium	Set up environment in Week 1, follow official installation guides, and assist each other to ensure uniform configuration.

6. Success Criteria

- Correctly identifies at least 8 date varieties
- Functional mobile app with all core features
- Smooth AI integration
- Completed and ready for final demo

Nakhlah

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